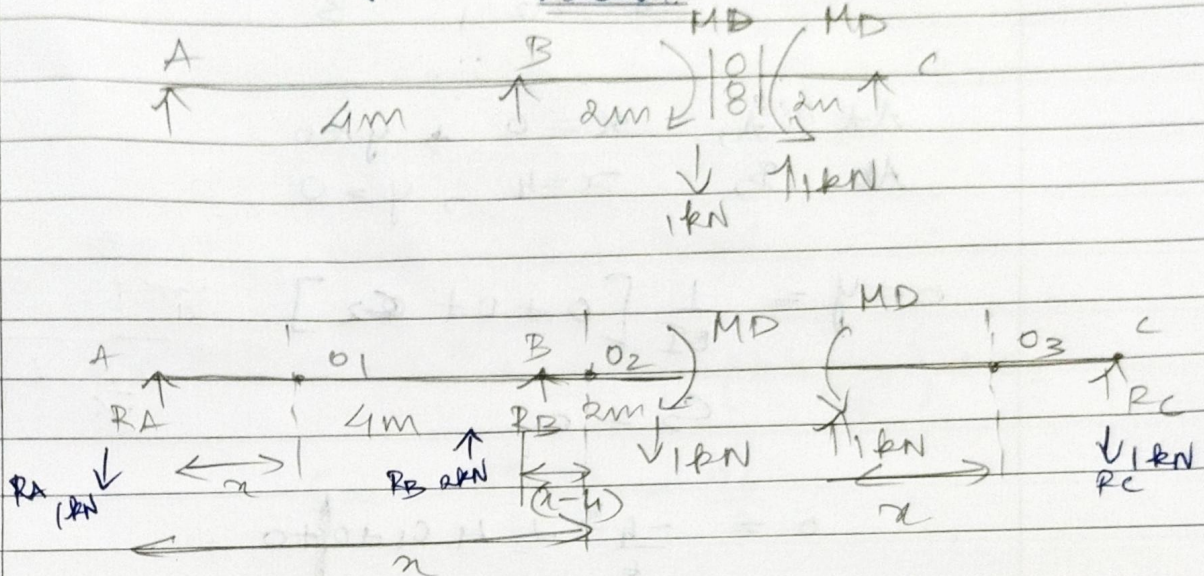


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Assignment-1
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ILD for SFD (from left side)

Take moment about C in DC

$$-MD + (1 \times 2) = 0$$

$$\Rightarrow MD = 2 \text{ kNm}$$

$$\Rightarrow RC = -1 \text{ kN}$$

Take moment about A in AD

$$MD - (RB \times 4) + (1 \times 6) = 0$$

$$\Rightarrow RB = 2 \text{ kN}$$

$$1 - RB = RA = 1 - 2 = -1 \text{ kN}$$

segment AD

$$\int EI \frac{d^2 y}{dx^2} = \int -1 \times x \quad ; \quad 2(x-4)$$

 $\Rightarrow$ slope eqn.

$$EI \frac{dy}{dx} = \frac{-x^2}{2} + C_1 \quad ; \quad \frac{2 \times (x-4)^2}{2}$$

$$EI y = \frac{-x^3}{6} + C_1 x \Big| \frac{(x-4)^3}{3} \Rightarrow \text{deflection eqn}$$

$$\text{At A, } x=0 \rightarrow y=0$$

$$\text{At B, } x=4 \rightarrow y=0$$

$$0=y = \frac{1}{EI} [0+0+C_2]$$

$$C_2 = 0$$

$$0 = \frac{-4^3}{6} + 4 C_1 + 0 + 0$$

$$C_1 = 8/3$$

$$y_{DX} = \frac{1}{EI} \left[\frac{-x^3}{6} + \frac{8x}{3} \Big| \frac{(x-4)^3}{3} \right]$$

$$y_{DA} \Big|_{x=6} = \frac{1}{EI} \left[\frac{-6^3}{6} + \frac{8 \times 6}{3} \Big| \frac{2^3}{3} \right] = \frac{-52}{3EI}$$

$$\theta_{DA} \Big|_{x=6} = \frac{1}{EI} \left[\frac{-6^2}{2} + \frac{8}{3} \Big| \frac{(6-4)^2}{2} \right]$$

$$\theta_{DA} = \frac{-11.33}{EI} = \frac{-34}{3EI}$$

Segment DC

$$\int EI \frac{d^2 y}{dx^2} = \int -2 + (1 \times x)$$

$$\int EI \frac{dy}{dx} = -2x + \frac{x^2}{2} + C_1$$

$$EI y = -\cancel{2} \frac{x^2}{2} + \frac{x^3}{6} + C_1 x + C_2$$

At C, $x=2, y=0$

At D, $0 \text{ DA} - 0 \text{ DC } x=0$

$$EI (0) = -2^2 + \frac{2^3}{6} + \left(2x - \frac{34}{3}\right) + C_2$$

$$C_2 = \frac{25.33}{EI} = \frac{76}{3}$$

$$\frac{EI(-11.33)}{EI} = 0 + 0 + C_1$$

$$\Rightarrow C_1 = \frac{-34}{3}$$

$$y_{DC} = -x^2 + \frac{x^3}{6} - \frac{34}{3}x + \frac{76}{3}$$

$$0 \text{ DC} = \frac{-34}{3EI}$$

$$y_{DC} \Big|_{x=0} = \frac{1}{EI} \left[0 + 0 + 0 + \frac{76}{3} \right] = \frac{76}{3EI}$$

$$y_{DD} = \frac{1}{EI} \left[\frac{76}{3} + \frac{52}{3} \right] = \frac{128}{3EI}$$

segment AD

segment DC

$$y_{xx} = \frac{1}{EI} \left[\frac{-x^3}{6} + \frac{8x}{3} - \frac{(x-4)^3}{3} \right]$$

$$\frac{1}{EI} (128/3)$$

$$y_{xx} = \frac{1}{EI} \left[-\frac{x^2}{6} + \frac{x^3}{3} - \frac{34x}{3} + \frac{76}{3} \right]$$

$$\frac{1}{EI} (128/3)$$

x	0	1	2	3	4	5	6
TIP	0	0.059	0.094	0.082	0	-0.168	-0.406

	0	1	2
	0.594	0.309	0

BMD

